□ Session 17: Exploit Analysis Automation & Debugging Basics

□ Part 1: Exploit Analysis Automation Process

□ What is Exploit Analysis?

Exploit analysis is the process of **examining how a vulnerability is exploited** to:

- Understand the nature of the exploit
- Determine the risk/severity
- Recreate and automate testing or patching processes

⚙️ Why Automate Exploit Analysis?

Manual analysis is slow and prone to error. Automation helps in:

- **Efficient detection** of known exploits
- **Repetitive testing** using fuzzers
- **Scalable scanning** across systems
- **Report generation** for vulnerabilities

🛠️ Automation Tools & Techniques

Tool	Purpose

Metasploit	Exploit testing, payload delivery
Boofuzz	Fuzzing automation
pwntools	Scripting exploitation in Python
Ghidra/IDA	Reverse engineering with scripting
AutoSploit	Mass-exploit known CVEs
Nmap NSE	Scriptable exploit detection

Process Workflow

1. **Input collection** (target IPs, binaries)

2. **Vulnerability scanning** (e.g., Nmap, Nessus)

3. **Auto-exploitation** (Metasploit, AutoSploit)

4. **Logging/reporting**

5. **Custom scripting** using Python or bash

□ Sample Python Script Using pwntools

python

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from pwn import * p = process('./vuln_app') payload = b'A' * 64 + b'\x90\x90...' p.sendline(payload) p.interactive()

□ Part 2: Debugging Basics

🐞 What is Debugging?

Debugging is the **process of identifying, isolating, and fixing bugs** in software or systems.

🔍 Types of Debugging

- **Static Debugging**: Without running code (code review)
- **Dynamic Debugging**: During runtime (using debugger)

🧠 Basic Debugging Concepts

Term	Description
Breakpoint	Stops execution at a given line
Step Over	Executes current line and moves ahead

Step Into	Jumps inside function call
Watch Variable	Monitor variable value at runtime
Stack Trace	Function call history

Tools

- **GDB** – GNU Debugger (Linux)
- **PDB** – Python Debugger
- **OllyDbg** – Windows binary debugger
- **x64dbg** – Windows x64 debugger

- **VSCode** – For Python/JavaScript/etc.

□ Example: Debugging Python Code

python

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import pdb def divide(x, y): pdb.set_trace() return x / y divide(10, 0)

Output:

scss

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(Pdb) l (Pdb) p x (Pdb) p y (Pdb) n
